

Activity-4

Given input {4371, 1323, 6173, 4199, 9679, 1989} and a hash function $h(n) = n \pmod{10}$, show the result.

a) Separate chaining hash table.

hash fn:

$$h(n) = n \pmod{10}$$

hash values:

key	$h(n)$
4371	1
1323	3
6173	3
4199	9
4344	4
9679	9
1989	9

Result:

index chain

0

-

1

4371

2

-

3

1323-6173

4

4344

5

-

6

-

7

-

8

-

9

4199 → 9679 → 1989

(P) linear probing:

formula:

$$h(n) = (h(n) + i) \% 10$$

insertions:

4371 $\rightarrow$ 1

1323 $\rightarrow$ 3

6173 $\rightarrow$ 3 $\rightarrow$ 4

4199 $\rightarrow$ 9

4344 $\rightarrow$ 4 $\rightarrow$ 5

9679 $\rightarrow$ 9 $\rightarrow$ 0

1989 $\rightarrow$ 9 $\rightarrow$ 0 $\rightarrow$ 1

Results

index	Value
0	9679
1	4371
2	1989
3	1323
4	6173
5	4344
6	-
7	-
8	-
9	4199

(P) Quadratic probing:

$$h(n) = (h(n) + i^2) \% 10$$

insertions:

4371 $\rightarrow$ 1

1323 $\rightarrow$ 3

6173 $\rightarrow$ 3 $\rightarrow$ 4

4199 $\rightarrow$ 94344 $\rightarrow$ 4 $\rightarrow$ 59679 $\rightarrow$ 9 $\rightarrow$ 01989 $\rightarrow$ 9 $\rightarrow$ 0 $\rightarrow$ 3 $\rightarrow$ 8Result:

index	Value
0	9679
1	4371
2	-
3	1323
4	6173
5	4344
6	-
7	-
8	1989
9	4199

d) Double-hashing

$$h_1(n) = n \% 10$$

$$h_2(n) = 7 - (n \% 7)$$

$$h_3(n) = (h_1(n) + p \times h_2(n)) \% 10$$

 $\Rightarrow$ The second hash values are

Key	$h_1(n)$	$h_2(n)$
4371	1	4
1323	3	7
6173	3	1
4199	9	1
4344	4	3

9679

9

2

1989

9

6

final table :

index	value
0	-
1	4371
2	-
3	1323
4	6173
5	9679
6	-
7	4344
8	-
9	4199

~~1989~~ cannot be inserted (cycle detected)